mcpy: Canonical & Grand Canonical Monte Carlo for atomistic systems with machine-learning potentials

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Summary

mcpy is a Python package for Monte Carlo (MC) simulations of atomistic systems (surfaces, nanoparticles, fluids, and bulk) at controlled temperature, volume, and particle number or chemical potential. It is built on the Atomic Simulation Environment (ASE) (Larsen et al., 2017), so any ASE-compatible calculator can drive the sampling: density-functional theory codes, classical force fields, or machine-learning interatomic potentials (MLIPs) such as MACE (Batatia et al., 2022). mcpy implements canonical and grand canonical ensembles, as well as replica-exchange approaches (Swendsen & Wang, 1986), in modular run loops and runs with or without local relaxation of trial moves. Following the hybrid scheme of Senftle et al. (2014), insertions and deletions can be relaxed locally before the acceptance test, which makes sampling efficient in grand canonical ensemble simulations of densely packed metallic systems where rigid insertions would almost always be rejected; relaxation can equally be switched off for standard MC of fluids and other systems. The same loop runs on CPUs via MPI or on GPUs through different backends such as NVIDIA Alchemi, which relaxes a batch of replicas in a single evaluation. The package provides modular trial moves (insertion, deletion, displacement, permutation (enabling chemical-ordering optimization in multi-component systems), shake, and short MD runs via Brownian/Langevin moves), configurable cell geometries (periodic box, rectangular sub-slab, spherical region around a nanoparticle, and user-defined cells), global optimization and statistical sampling, together with post-processing utilities that turn raw ensembles into surface and nanoparticle phase diagrams.

Statement of need

Predicting the equilibrium structure and composition of surfaces and nanoparticles under reactive atmospheres requires sampling over candidate structures and compositions (i.e. particle number) in the grand canonical ensemble at near–first-principles accuracy. MLIPs now deliver close-to-DFT accuracy at a small fraction of the cost, making large-scale GCMC of realistic structures increasingly practical. Yet no open-source package combines relaxation-coupled GCMC, replica exchange, an ASE-native MLIP backend, and the cell geometries needed for finite nanoparticles in a single, modular workflow.

mcpy's grand canonical simulations are aimed at computational chemists and materials scientists studying heterogeneous catalysis, oxidation, and gas–surface equilibria, where the stable composition, not just a fixed structure, is the quantity of interest. Although it was developed for these applications, the grand canonical sampling is agnostic to the level of theory or the targeted system: it runs on any system ASE can represent and any ASE-compatible calculator, from molecular fluids to bulk solids. The same workflow therefore extends well beyond the metallic surfaces and nanoparticles that motivated it.

State of the field

Several established codes perform Monte Carlo sampling of atomistic systems, but each covers only part of the space `mcpy` targets. RASPA2 (Dubbeldam et al., 2016) and its recent C++ rewrite RASPA3 (Ran et al., 2024) are specialized for adsorption and diffusion of molecules in nanoporous materials with classical force fields. DL_MONTE (Brukhno et al., 2021) is a general-purpose Monte Carlo code that supports GCMC of solids and fluids, but with classical interaction models and its own input ecosystem rather than native integration with MLIPs or ASE. LAMMPS (Thompson et al., 2022) provides a GCMC fix, parallel tempering, and machine-learning pair styles, but it neither couples replica exchange to grand canonical sampling nor exposes a relaxation-coupled GCMC in which an arbitrary ASE calculator drives each acceptance test. ASE itself (Larsen et al., 2017) supplies the `Atoms` object and a unified calculator interface spanning DFT codes, classical force fields, and MLIPs, but no grand canonical ensemble built on top of it.

We built `mcpy` as a new package rather than extending any one of these because the capabilities it unifies live in different ecosystems: the calculator-agnostic accuracy of ASE, the grand canonical and replica-exchange machinery of dedicated Monte Carlo codes, and the geometric flexibility needed for finite nanoparticles. Adding MLIP-native, relaxation-coupled GCMC to RASPA2 or DL_MONTE would require reimplementing ASE's calculator abstraction, while wrapping LAMMPS in a relaxation-coupled, ASE-native grand canonical loop with free-volume cell geometries and *ab initio* thermodynamics post-processing would reproduce much of `mcpy`'s MC design around an MD-centric engine. Building directly on ASE instead lets `mcpy` inherit the entire ASE calculator ecosystem and contribute the missing grand canonical layer. That layer is the package's distinct contribution: ASE-native, MLIP-driven, relaxation-coupled GCMC with replica exchange and nanoparticle/surface geometries.

Software design

`mcpy` is organized around four extensible abstractions: *ensembles* that drive the Monte Carlo loop (`GrandCanonicalEnsemble`, `CanonicalEnsemble`, `ReplicaExchange`), *moves* that propose trial configurations (insertion, deletion, displacement, permutation, shake, and Brownian), *cells* that define the sampling region and estimate its free volume (periodic box, rectangular sub-slab, spherical region around a nanoparticle, and user-defined cells), and *calculators* that wrap ASE energy evaluations. This separation reflects the central design choice: by expressing the physics through ASE's `Atoms` and calculator interfaces, `mcpy` trades the raw speed of a monolithic, hand-tuned Monte Carlo kernel for the ability to drive identical sampling code with any ASE-compatible backend, from a DFT code to a MACE potential (Batatia et al., 2022). Because the dominant cost in the target application is the energy evaluation rather than the Monte Carlo bookkeeping, that trade-off is strongly favorable, and it lets a researcher exchange accuracy for cost without rewriting the simulation.

The second design choice is an optional relaxation-coupled acceptance criterion. Following the hybrid scheme of Senftle et al. (2014), each trial insertion or deletion can be relaxed locally before the acceptance test. This adds cost to each step, but in densely packed metallic surfaces and nanoparticles a rigid insertion overlaps neighboring atoms and is almost always rejected; the local relaxation is what makes grand canonical sampling converge at all for these systems, so the per-step cost is the enabling trade-off rather than an overhead. Because the relaxation is delegated to ASE, any of ASE's local-relaxation algorithms can be selected according to its suitability for the chosen system and level of theory. Conversely, the relaxation step can be disabled entirely, recovering standard unbiased Metropolis Monte Carlo that samples the exact Boltzmann distribution, appropriate for fluids and other systems where rigid trial moves are readily accepted.

The third choice concerns ergodicity and parallelism. Replica exchange (Swendsen & Wang, 1986) across temperatures or chemical potentials improves mixing, and `mcpy` exposes two complementary parallelization paths: over MPI ranks on CPUs, one replica per rank, or on a GPU through the NVIDIA Alchemi backend, which relaxes a batch of replicas together in a single evaluation. The MPI path is portable and dependency-light (`mpi4py` is optional); the GPU path trades that portability for throughput on large systems where batched relaxation dominates. Because the ensemble loop is decoupled from the calculator, the same replica-exchange code runs under either path.

Finally, the modular loop is reused beyond equilibrium sampling: every ensemble doubles as a basin-hopping global optimizer, emitting a running trajectory of strictly improving (minimum-energy) configurations, and replica-exchange runs additionally report the global minimum found across replicas. Compound moves that perturb several atoms per trial make these basin-hopping escapes effective in densely packed systems. Simulations log per-step energies, particle counts, and per-move acceptance ratios, and write extended-XYZ trajectories alongside an optional `minima` trajectory of strictly improving structures. The bundled phase-diagram utilities map sampled ensembles onto *ab initio* thermodynamics phase diagrams (Reuter & Scheffler, 2001), connecting the simulated chemical potential to experimental temperature and partial pressure.

To validate the grand canonical sampling independently of any MLIP, we reproduced two Lennard-Jones reference systems against LAMMPS (Thompson et al., 2022) (Figure 1): a purely Lennard-Jones fluid in reduced units and real argon in physical (eV) units. The average particle number and density as a function of chemical potential agree between the two codes across both unit systems, including the sharp gas–liquid condensation of argon. Although `mcpy` has so far been applied mainly to metal surfaces and nanoparticles, this benchmark confirms that its grand canonical machinery is correct and general, applicable to atomistic materials beyond the metallic systems that motivated it.

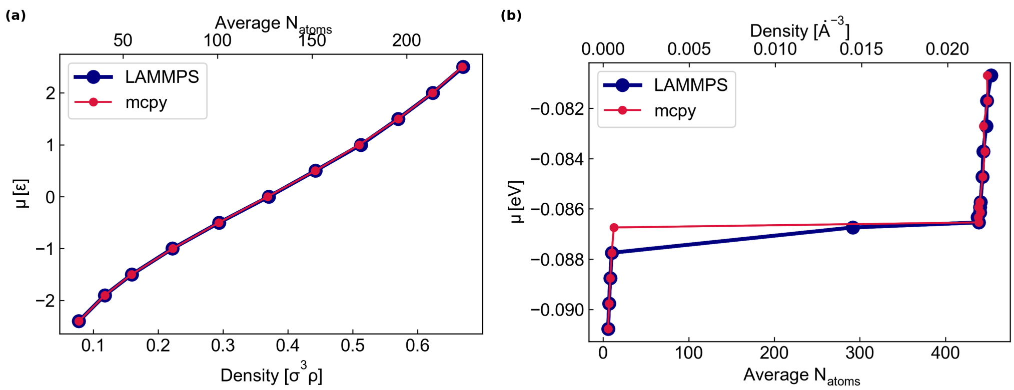

Figure 1: Validation of `mcpy`'s grand canonical sampling against LAMMPS (Thompson et al., 2022) for two Lennard-Jones reference systems. (a) A purely repulsive Lennard-Jones fluid in reduced units: chemical potential μ (in units of ϵ) versus reduced density $\sigma^3\rho$ and average particle number. (b) Real argon in physical units: μ (eV) versus average particle number and number density, capturing the sharp gas–liquid condensation. `mcpy` (red) reproduces LAMMPS (blue) across both unit systems.

Research impact statement

`mcpy` was developed to study the equilibrium oxidation of metal nanoparticles and surfaces under reactive atmospheres at near-DFT accuracy. The replica-exchange GCMC workflow was built during doctoral research at the Universitat de Barcelona (Farris, 2025) and underpins

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a forthcoming manuscript on the oxidation thermodynamics of silver nanoparticles, with further application to additional catalytic metal/oxide systems. Simulations run with `mcpy` have contributed to the group's grand canonical modelling of nanostructured catalysts under reaction conditions, presented at recent international conferences on cluster science and on machine learning for materials (Bruix, 2026; Telari et al., 2026). Figure 2 shows a representative result from this workflow: the hydrogenation phase diagram of a 201-atom PdAg nanoparticle, in which `mcpy` recovers the progressive hydrogen loading of the particle as the chemical potential increases. By coupling MLIP-driven energetics with grand canonical sampling and *ab initio* thermodynamics post-processing, `mcpy` predicts stable surface and nanoparticle compositions as a function of temperature and chemical potential, quantities that are otherwise inaccessible to fixed-composition relaxations. The package is open-source, documented with worked examples covering GCMC, replica exchange, and phase-diagram construction, and continuously tested across Python 3.11–3.13, so that the published workflows can be reproduced and extended by other groups.

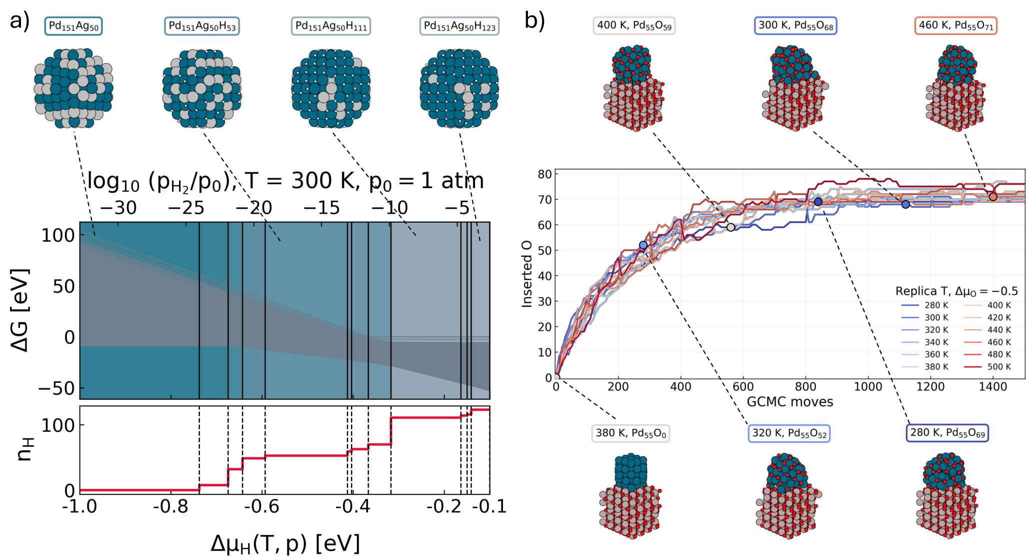

Figure 2: (a) Hydrogenation phase diagram of a 201-atom palladium–silver nanoparticle ($Pd_{151}Ag_{50}$) obtained with replica-exchange GCMC in `mcpy`, driven by the small AGNESI MACE potential (Batatia et al., 2025). The free energy of formation ΔG (in eV) is shown as a function of the hydrogen chemical potential $\Delta\mu_{\mathrm{H}}$ (equivalently $\log_{10}(p_{\mathrm{H}}/p_0)$ at $T = 300$ K); the insets show some of the most stable $Pd_{151}Ag_{50}H_z$ structures sampled for different chemical-potential windows, from the bare particle at low μ_{H} to a hydrogen-loaded $Pd_{151}Ag_{50}H_{123}$ at high pressure, highlighting the change in chemical ordering induced by the increase in hydrogen coverage. The chemical ordering of the bare $Pd_{151}Ag_{50}$ particle was first optimized with a single basin-hopping run; hydrogenated structures were then sampled by replica-exchange GCMC at hydrogen chemical potentials from $\Delta\mu_{\mathrm{H}} = -1$ to -0.35 eV with six temperatures per chemical-potential value, using H insertion, H deletion, and Pd-Ag swap moves. (b) Evolution of the number of inserted oxygen atoms during a replica-exchange GCMC simulation of a Pd_{55} fcc nanoparticle supported on a Al_2O_3 surface. The run consisted of 12 replicas with different temperatures at a fixed $\Delta\mu_{\mathrm{O}} = -0.5$. The structural insets show representative configurations sampled along the trajectories of the different replicas.

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AI usage disclosure

Claude (Opus) assisted with code refactoring, test scaffolding, packaging configuration, and editing of the manuscript and documentation. All AI-assisted outputs were reviewed, edited, and validated by the human authors, who designed the overall code architecture and take full responsibility for the accuracy and originality of the software and all submitted materials.

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